

50.057 Propositional Logic

ISTD, SUTD

Learning Outcomes

- ▶ Analyse the syntax of propositional statements
- ▶ Apply truth-table to analyse the semantics of the propositional statements
- ▶ Evaluate propositional statements
- ▶ Perform satisfaction check given propositional statements
- ▶ Explain logical equivalence
- ▶ Explain logical consistency
- ▶ Explain logical entailment
- ▶ Apply inference rules to devise propositional proofs

Propositional Logic

Propositional Logic is to reason about the connection between propositions. Each proposition is a statement denoting something is true or false.

From this point onwards, we use the terms “proposition”, “statement” and “sentence” interchangeably.

Two Kinds of Sentences

- ▶ A **simple sentence** formed by an atomic symbol, such as p , q , $it_is_raining$.
 - ▶ as a convention, we use a sequence of lowercase alphabets to denote an atomic symbol.
- ▶ A **complex sentence** constructed from a set of simple sentences connected by logical connectives, e.g.
 - ▶ $rain \wedge thunder$,
 - ▶ $bitter \vee sweet$,
 - ▶ $boy_older_than_18 \implies boy_goes_to_NS$,
 - ▶ $\neg floor_is_wet \implies \neg it_has_rained$.

Syntax of Propositional Logic

1. Conjunction $p \wedge q$
2. Disjunction $p \vee q$
3. Negation $\neg p$
4. Implication $p \implies q$
5. Biconditional $p \iff q$, a short hand for $(p \implies q) \wedge (q \implies p)$.

The order of precedence among the operators is arranged from strongest to weakest.

Precedence

1. $\neg$
2. $\wedge$
3. $\vee$
4. $\implies$
5. $\iff$

$\wedge$ and $\vee$ are **left** associative; $\implies$ and $\iff$ are **right** associative.

Precedence in Action

1. $\neg p \vee q$ is parsed as $(\neg p) \vee q$.
2. $p \wedge q \wedge r$ is parsed as $(p \wedge q) \wedge r$.
3. $p \implies q \implies r$ is parsed as $p \implies (q \implies r)$.
4. $p \implies q \wedge q \implies p$ would be parsed as $p \implies ((q \wedge q) \implies p)$, which is quite different from $(p \implies q) \wedge (q \implies p)$.

Semantics of Propositional Logic

The meaning of propositional logic statements is represented as the relation between the truth assignment of the atomic propositions and the truth values of complex statements.

We have studied them in our earlier school years, under the name “**truth table**”.

Truth Table

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \implies q$
1	1	0	1	1	1
1	0	0	0	1	0
0	1	1	0	1	1
0	0	1	0	0	1

Without confusing ourselves with the propositions, we use 1 to denote true and 0 to denote false.

We leave the semantics of biconditional statements as an exercise.

Evaluation

- ▶ Determine the truth value of any compound statement given a truth-value assignment of all the atomic propositions.
- ▶ c.f. Boolean algebra
- ▶ For example, given p is 1 and q is 0, we find

$$\begin{aligned} p &\implies ((q \wedge q) \implies p) \\ p &\implies ((0 \wedge 0) \implies p) \\ p &\implies (0 \implies p) \\ 1 &\implies 1 \\ 1 \end{aligned}$$

Satisfaction

- ▶ Satisfaction is the opposite of evaluation.
- ▶ Given a complex statement ψ , find the set (or subset) of truth-assignments to the atomic propositions such that ψ is evaluated to 1.
- ▶ We could construct the truth table over all the atomic propositions in ψ and filter out those rows that make ψ evaluate to 0.
 - ▶ working but not scaling well

Satisfaction: An Example

For example given statement $(p \implies q) \wedge (q \implies p)$,

p	q	satisfying $(p \implies q) \wedge (q \implies p)$
1	1	yes
1	0	no
0	1	no
0	0	yes

It shows that the 1st and the 4th rows of truth-assignments satisfy $(p \implies q) \wedge (q \implies p)$.

Valid, Unsatisfiable, Contingent

A proposition is

- ▶ *unsatisfiable* if there exists no truth-assignment to make it evaluate to 1.
- ▶ *valid* if there exists no truth-assignment to make it evaluate to 0.
- ▶ *contingent* if it is in neither categories.
- ▶ *satisfiable* if it is valid or contingent.
- ▶ *falsifiable* if it is unsatisfiable or contingent.

Logical Equivalence and Logical Consistency

- ▶ ψ and ϕ are logically *equivalent* if for all truth-assignments, they yield the same evaluation results.
 - ▶ For example $\neg(p \vee q)$ and $\neg p \wedge \neg q$ are logically equivalent.
 - ▶ If ψ and ϕ are logically equivalent, we can use ψ in the context where ϕ is expected, which is called *substitution*.
- ▶ ψ and ϕ are logically *consistent*, if there exists some truth-assignment under which both ψ and ϕ are evaluated to 1.
- ▶ In the above we use ψ and ϕ instead of p and q to highlight that we are referring to any *general* proposition.

Logical Entailment (Formally)

Given two propositions, ψ and ϕ , if all the truth-assignments satisfying ψ also satisfy ϕ , we say ψ logically entails ϕ , written as $\psi \models \phi$.

For example, we find $p \models (p \vee q)$, but $p \not\models (p \wedge q)$.

Exercise

Verify $p \models (p \vee q)$ using truth-table.

Equivalence Theorem

ψ and ϕ are logically equivalent **if and only if** $\psi \iff \phi$ is valid.

This theorem enables us to rewrite larger (complex) sentences into alternative forms via substitution so that we could apply some other result to prove it.

Deduction Theorem

ψ logically entails ϕ **if and only if** $(\psi \implies \phi)$ is valid.

More generally, a finite set of sentences $\{\psi_1, \dots, \psi_n\}$ logically entails ϕ if and only if the compound sentence $(\psi_1 \wedge \dots \wedge \psi_n \implies \phi)$ is valid.

This theorem enables us to use logical implication to make “progress” towards the goal of the proof.

Unsatisfiability Theorem

A set of sentences, $\{\psi_1, \dots, \psi_n\}$ logically entails a sentence ϕ **if and only if** $\psi_1 \wedge \dots \wedge \psi_n \wedge (\neg\phi)$ is unsatisfiable.

This theorem unlocks one possible proof strategy for proving entailment, i.e. **proof by contradiction**.

Consistency Theorem

ϕ is logically consistent with ψ **if and only if** the sentence $(\psi \wedge \phi)$ is satisfiable.

More generally, a sentence ψ is logically consistent with a finite set of sentences $\{\psi_1, \dots, \psi_n\}$ if and only if the compound sentence $(\psi_1 \wedge \dots \wedge \psi_n \wedge \psi)$ is satisfiable.

Proof System: Natural Deduction

- ▶ The process of proving in propositional logic is to
 1. start from a set of premises (each premise is a proposition which is assumed to be true), and
 2. check that the given conclusion (which is our goal proposition) is entailed.
- ▶ What about using truth-table?
 - ▶ it won't scale when the set of premises and the conclusion is complicated.
- ▶ Alternatively, we can make use of *inference rules*.
 - ▶ These rules are not part of the propositional syntax. (Meta level)
 - ▶ Although we could also prove their validity by constructing the truth-tables, having them stated as inference rules speeds up our proof process (by shortening the proof).

One way to understand the inference rules is to think of them as the “builtin” library for our proof system.

Implication Elimination ($\implies$ Elim)

$$\frac{\phi \implies \psi \quad \phi}{\psi}$$

- ▶ In this rule, the two sentences above the horizontal line are the premises. The one below is called the conclusion.
 - ▶ Since the rule itself can also be proven valid, we could restore the rule back into an implication statement $(\phi \implies \psi) \wedge \phi \implies \psi$ and use truth-table to prove its validity.
 - ▶ This is just for understanding; we will apply this rule directly without going through its validity proof.
- ▶ When given a context containing $it_rain \implies the_floor_is_wet$ and it_rain , we could apply the implication elimination rule to show that $the_floor_is_wet$.

Implication Introduction ($\implies$ Intro)

$$\frac{\phi \vdash \psi}{\phi \implies \psi}$$

- ▶ This rule is to support “sub-proof”.
- ▶ The notation $\phi \vdash \psi$ means that there is a way we can verify ψ given ϕ . (i.e. the sub proof)
- ▶ The implication introduction rule discharges that temporary assumption and concludes that $\phi \implies \psi$ given a sub proof from ϕ to show ψ .

A more detailed example will be given later.

And Elimination ($\wedge$ Elim)

$$\frac{\phi_1 \wedge \phi_2 \wedge \dots \wedge \phi_n}{\phi_i}$$

This rule says that given a conjunction of predicates is true, then the individual predicate must be true.

For example, from

$$rain \wedge cold$$

we may derive *rain*, and we may also derive *cold*. The rule can select any one of the conjuncts in the larger conjunction.

And Introduction ($\wedge$ Intro)

$$\frac{\begin{array}{c} \phi_1 \\ \phi_2 \\ \dots \\ \phi_n \end{array}}{\phi_1 \wedge \phi_2 \wedge \dots \wedge \phi_n}$$

- ▶ This rule says that if a set of predicates is true, then the conjunction of them must be true.
 - ▶ For example, if we have separately established that it is raining and that the floor is wet, we may combine them to derive

$$rain \wedge floor_is_wet$$

- ▶ Both premises are required. Having only *rain* would not be enough to apply this rule and conclude *rain $\wedge$ floor_is_wet*.

Or Elimination ($\vee$ Elim)

$$\begin{array}{c} \phi_1 \vee \phi_2 \vee \dots \vee \phi_n \\ \phi_1 \implies \psi \\ \phi_2 \implies \psi \\ \dots \\ \phi_n \implies \psi \\ \hline \psi \end{array}$$

- ▶ This rule says that if a disjunction of predicates is true, and all the predicates in the disjunction lead to the same consequence, then the consequence must be true. For example,
 - ▶ If we know that either *tom_eats_all_the_candies* or *jane_eats_all_the_candies* is true.
 - ▶ And *tom_eats_all_the_candies* $\implies$ *no_candies_left* and *jane_eats_all_the_candies* $\implies$ *no_candies_left*.
 - ▶ We can conclude that *no_candies_left*.

Or Introduction ($\vee$ Intro)

$$\frac{\phi_i}{\phi_1 \vee \phi_2 \vee \dots \vee \phi_n}$$

- ▶ This rule says that once one alternative has been established, it is valid to state that at least one of several alternatives holds. The chosen conclusion may contain the known proposition in any position.
 - ▶ For example, from *rain* we may derive *rain* $\vee$ *snow*.
 - ▶ Similarly, from *snow* we may derive *rain* $\vee$ *snow*.
 - ▶ We do not need to prove that both alternatives are true; proving one of them is sufficient.

Negation Elimination ($\neg$ Elim)

$$\frac{\neg\neg\phi}{\phi}$$

- ▶ This rule removes two consecutive negations. For example given

$$\neg\neg rain$$

- ▶ we can verify *rain*.
- ▶ The rule does not allow us to infer *rain* from $\neg rain$; the premise must contain the double negation $\neg\neg rain$.

Negation Introduction ($\neg$ Intro)

$$\frac{\begin{array}{l} \phi \implies \psi \\ \phi \implies \neg\psi \end{array}}{\neg\phi}$$

- ▶ This rule is a **proof by contradiction**.
- ▶ If assuming ϕ leads both to ψ and to $\neg\psi$, then the assumption ϕ cannot be true.
- ▶ For example, if

$$p \implies q \quad \text{and} \quad p \implies \neg q,$$

- ▶ then we may conclude $\neg p$.

Biconditional Elimination ($\iff$ Elim)

$$\frac{\phi \iff \psi}{\begin{array}{l} \phi \implies \psi \\ \psi \implies \phi \end{array}}$$

- ▶ This rule extracts the two directional implications contained in a biconditional. For example, from

$$student_passes \iff credits_sufficient$$

- ▶ we may derive both
 - ▶ $student_passes \implies credits_sufficient$ and
 - ▶ $credits_sufficient \implies student_passes$.

Biconditional Introduction ($\iff$ Intro)

$$\frac{\begin{array}{l} \phi \implies \psi \\ \psi \implies \phi \end{array}}{\phi \iff \psi}$$

- ▶ This rule combines two implications into a biconditional. For example, if we have proved both

$$\text{even} \implies \text{divisible_by_two} \quad \text{and} \quad \text{divisible_by_two} \implies \text{even},$$

- ▶ then we may conclude $\text{even} \iff \text{divisible_by_two}$.

An Example Proof

Suppose given

$$\{p, (p \implies q), ((p \implies q) \implies (q \implies r))\}$$

as the set of premises, we want to show that r is the conclusion.

1.	p	Premise
2.	$p \implies q$	Premise
3.	$((p \implies q) \implies (q \implies r))$	Premise
4.	q	$\implies$ Elim : 2, 1
5.	$q \implies r$	$\implies$ Elim : 3, 2
6.	r	$\implies$ Elim : 5, 4

Provability

Let

- ▶ R be a set of inference rules,
- ▶ Δ be a set of propositional premises, and
- ▶ ϕ denotes a conclusion.

We write $\Delta \vdash_R \phi$ iff there exists a sequence of rewriting steps that leads us to ϕ starting from Δ .

Sometimes we omit the subscript R to just write $\Delta \vdash \phi$.

Sub-proof (Conditional Proof)

Given

$$\{p, p \implies q, q \implies r\}$$

we want to show r . In the absence of the implication transitivity rule, we can develop the proof as follows.

Sub-proof (Cont'd)

1.	p	Premise
2.	$p \implies q$	Premise
3.	$q \implies r$	Premise
4.1.	p	Reiterate : 1
4.2.	q	$\implies$ Elim : 2, 4.1
4.3.	r	$\implies$ Elim : 3, 4.2
4.	$p \implies r$	$\implies$ Intro : 4.1 – 4.3
5.	r	$\implies$ Elim : 1, 4

In the above proof, steps 4.1 - 4.3 form a sub proof $p \vdash r$ (i.e. p entails r). We can use the rule ($\implies$ Intro) to create an implication $p \implies r$ into our proof context.

One may argue that the above proof can be established without sub-proof as we already have r in 4.3. The above example is to show the **possibility** of using the ($\implies$ Intro), not the necessity. In many situations sub-proof is useful, i.e., we might need to reuse the fact that $p \implies r$ multiple times to prove something else.

Soundness

Our proof system stated above is **sound**.

Let Δ be a set of propositional premises, and ϕ denotes a conclusion, we have $\Delta \vdash \phi$ implies $\Delta \models \phi$.

The meaning of the above statement is that if there exists a proof that brings us from Δ to ϕ , then Δ entails ϕ (can be verified using truth-table).

Completeness

Our proof system stated above is **complete**.

Let Δ be a set of propositional premises, and ϕ denotes a conclusion, we have $\Delta \models \phi$ implies $\Delta \vdash \phi$.

The meaning of the above statement is that if Δ entails ϕ (can be verified using truth-table), then there exists a proof that brings us from Δ to ϕ .

Other Proof Systems

- ▶ Hilbert System: make our propositional proof less structural, hence more suitable for proving meta theory;
- ▶ Normalization / CNF Transformation: canonicalize the premises and intermediate propositions.

Further Readings

- ▶ http://intrologic.stanford.edu/chapters/chapter_04.html
- ▶ http://intrologic.stanford.edu/chapters/chapter_05.html (what we covered)
- ▶ http://intrologic.stanford.edu/chapters/chapter_06.html